

Solution 8

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BS24B006

all csvs and python file are attached here

https://github.com/specapoorv/BT3040_BioInfo_Lab_assignments/tree/main/assignment_8

q.1

assignment_8 > q1.py > ...

```
1 import matplotlib.pyplot as plt
2
3 # hydrophobicity values given
4 hydro_scale = {
5     "A": 13.85,
6     "D": 11.61,
7     "C": 15.37,
8     "E": 11.38,
9     "F": 13.93,
10    "G": 13.34,
11    "H": 13.82,
12    "I": 15.28,
13    "K": 11.58,
14    "L": 14.13,
15    "M": 13.86,
16    "N": 13.02,
17    "P": 12.35,
18    "Q": 12.61,
19    "R": 13.10,
20    "S": 13.39,
21    "T": 12.70,
22    "V": 14.56,
23    "W": 15.48,
24    "Y": 13.88,
25 }
26
27 # sequences from fasta
28 s1 = "FDCAEYRSTNIYGYGLYEVSMPAKNTGIVSSFFTYTGPAGHTQWEIDIEFLGKDTTKVQFNYYTNGVGGHEKVISLGFDASKGFHTYAFDWQ
29 s2 = "KASEDLVKKHAGVLGAILKKKGHHEALKPLAQSHATKAHKNIFISEAIIHVLHSRHPGDFGADAQGAMNKALELFRKDIAAKYKELGY"
30 s3 = "TVEGAGSIAAATGFVKDQLGKNEEGAPQEGILEDMPVDPDNEAYEMPSEEGYQDYEPEA"
31
32 seqs = [("seq1", s1), ("seq2", s2), ("seq3", s3)]
33
34
35 # simple sliding window function
36 def get_profile(seq, win):
37     scores = []
38     positions = []
39     half = win // 2
40     for i in range(len(seq) - win + 1):
41         window_seq = seq[i : i + win]
42         total = 0
43         for aa in window_seq:
44             total = total + hydro_scale[aa]
45         avg = total / win
46         scores.append(avg)
47         positions.append(i + half + 1)
48     return positions, scores
```

main*   Run Testcases  0  0  Live Share

Spaces: 4 UTF-8 LF {} Python Python 3.10.12



```

49
50
51 # secondary structure scanning:
52 # window 5-7 for beta sheets, 9-11 or 19 for alpha helices
53 # using window 7 for beta and 11 for alpha
54 for name, s in seqs:
55     print("=" * 40)
56     print("Sequence:", name)
57     print("Length:", len(s))
58
59     # calculate profiles
60     pos_beta, prof_beta = get_profile(s, 7)
61     pos_alpha, prof_alpha = get_profile(s, 11)
62
63     # find regions above average hydrophobicity threshold (~13.6)
64     threshold = 13.6
65
66     print("Predicted Beta-strand candidate centers (window=7, cutoff > 13.6):")
67     beta_peaks = []
68     for p, score in zip(pos_beta, prof_beta):
69         if score > threshold:
70             beta_peaks.append((p, round(score, 2)))
71     print(beta_peaks)
72
73     print("Predicted Alpha-helix candidate centers (window=11, cutoff > 13.6):")
74     alpha_peaks = []
75     for p, score in zip(pos_alpha, prof_alpha):
76         if score > threshold:
77             alpha_peaks.append((p, round(score, 2)))
78     print(alpha_peaks)
79
80     # plot
81     plt.figure(figsize=(10, 4))
82     plt.plot(pos_beta, prof_beta, label="Beta window (w=7)", color="blue")
83     plt.plot(pos_alpha, prof_alpha, label="Alpha window (w=11)", color="red")
84     plt.axhline(
85         y=threshold, color="gray", linestyle="--", label="Hydrophobic cutoff"
86     )
87     plt.title(f"Hydrophobicity Profile - {name}")
88     plt.xlabel("Residue Position")
89     plt.ylabel("Mean Hydrophobicity")
90     plt.legend()
91     plt.tight_layout()
92     plt.show()

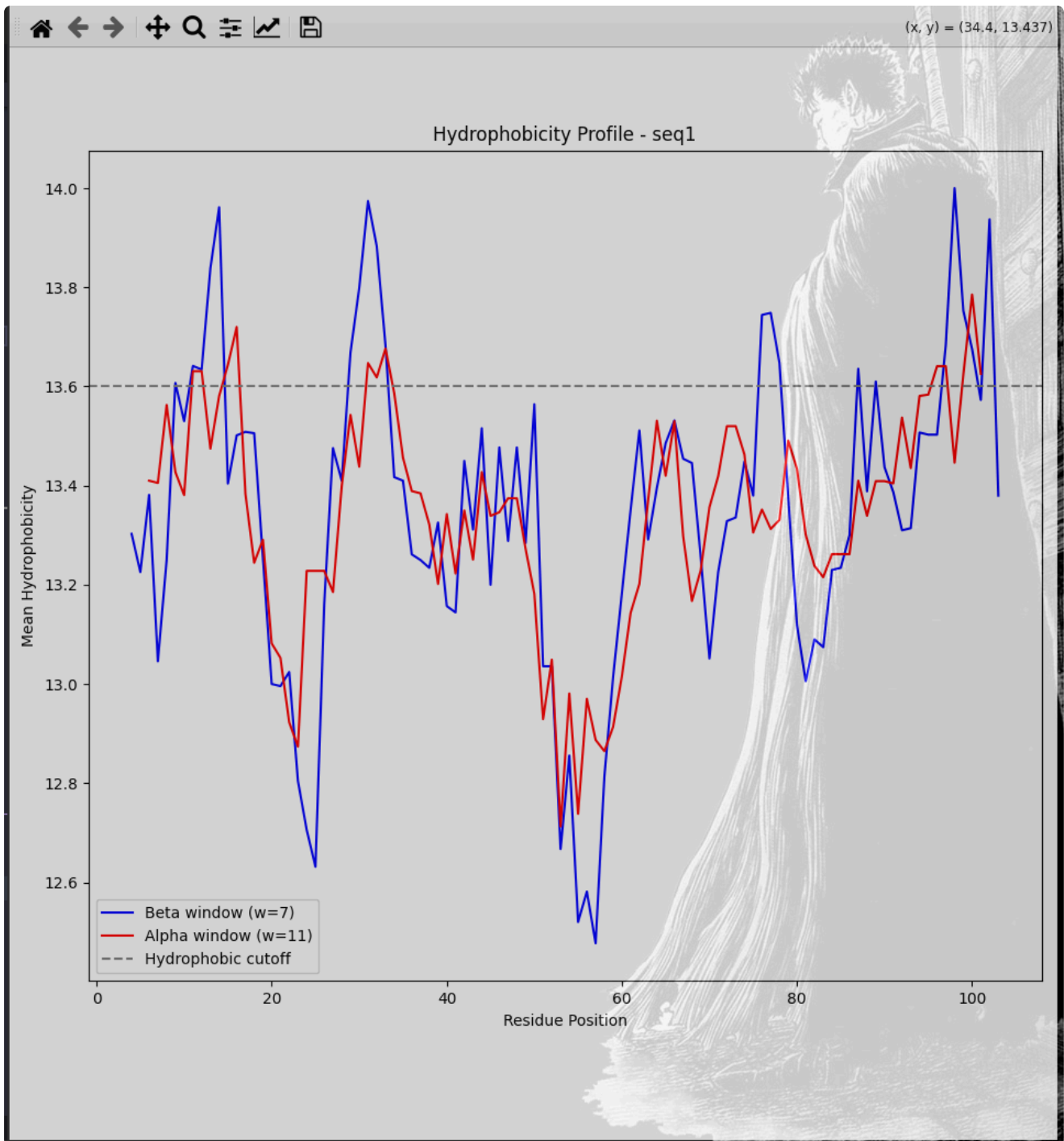
```

results

```

Sequence: seq1
Length: 106
Predicted Beta-strand candidate centers (window=7, cutoff > 13.6):
[(9, 13.61), (11, 13.64), (12, 13.63), (13, 13.84), (14, 13.96), (29, 13.67), (30, 13.8), (31, 13.97),
(32, 13.88), (33, 13.68), (76, 13.74), (77, 13.75), (78, 13.65), (87, 13.64), (89, 13.61), (97, 13.68),
(98, 14.0), (99, 13.75), (100, 13.68), (102, 13.94)]
Predicted Alpha-helix candidate centers (window=11, cutoff > 13.6):
[(11, 13.63), (12, 13.63), (15, 13.64), (16, 13.72), (31, 13.65), (32, 13.62), (33, 13.68), (96, 13.64),
(97, 13.64), (99, 13.62), (100, 13.79), (101, 13.63)]

```

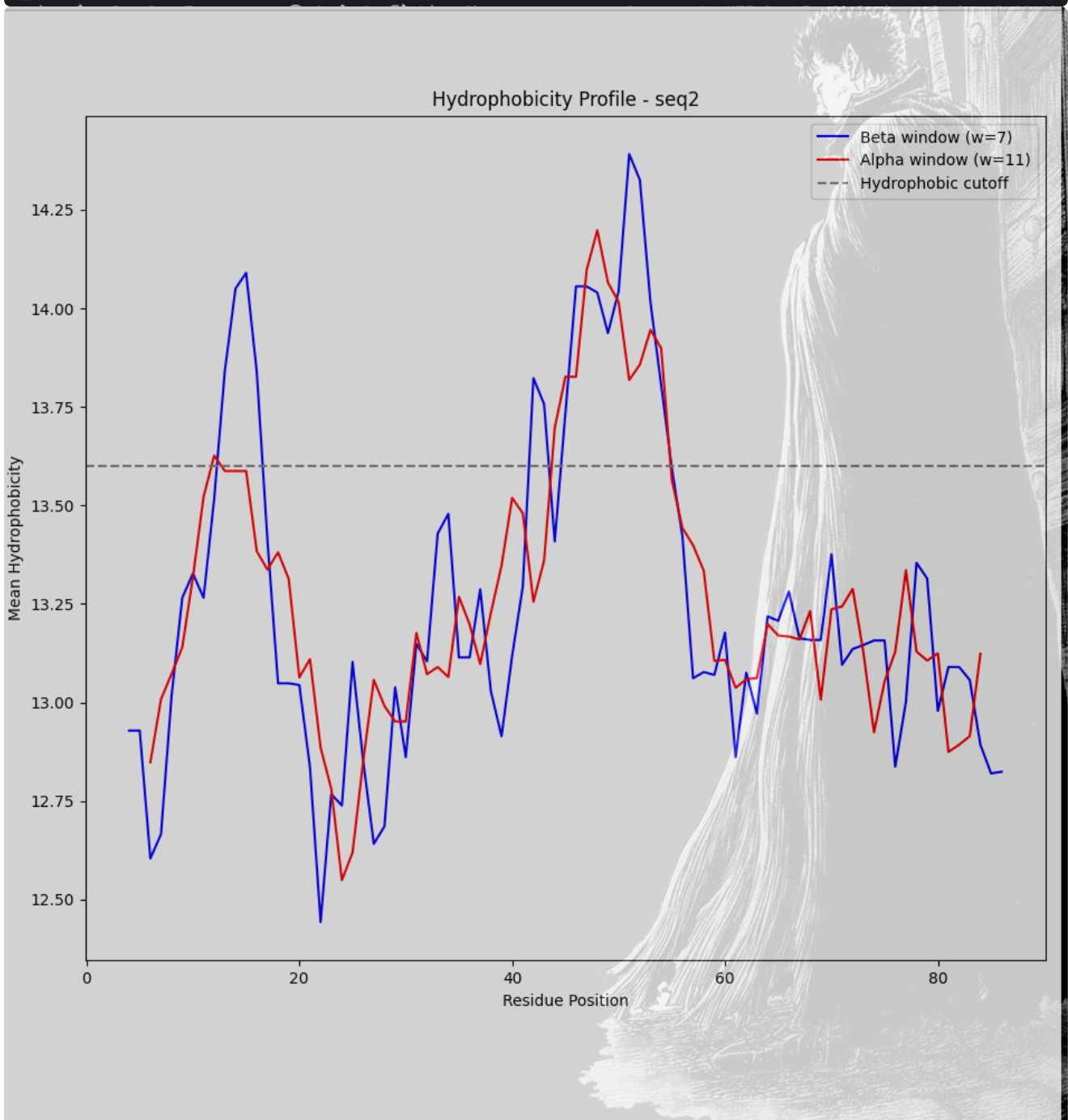


seq1 is mostly beta sheets because the plot has sharp narrow peaks instead of broad plateaus. beta strands are found around residues 9-14, 29-33, 76-78, and 97-102 where values cross 13.6. the alpha window crosses cutoff at the same spots but only briefly, so they are just hydrophobic beta cores rather than real helices. the deep dips are hydrophilic loops between the strands.

```

Sequence: seq2
Length: 89
Predicted Beta-strand candidate centers (window=7, cutoff > 13.6):
[(13, 13.84), (14, 14.05), (15, 14.09), (16, 13.84), (42, 13.82), (43, 13.76), (45, 13.73), (46, 14.06),
, (47, 14.06), (48, 14.04), (49, 13.94), (50, 14.04), (51, 14.39), (52, 14.33), (53, 14.01), (54, 13.81
)]
Predicted Alpha-helix candidate centers (window=11, cutoff > 13.6):
[(12, 13.63), (44, 13.7), (45, 13.83), (46, 13.83), (47, 14.1), (48, 14.2), (49, 14.07), (50, 14.02), (
51, 13.82), (52, 13.86), (53, 13.95), (54, 13.9)]

```

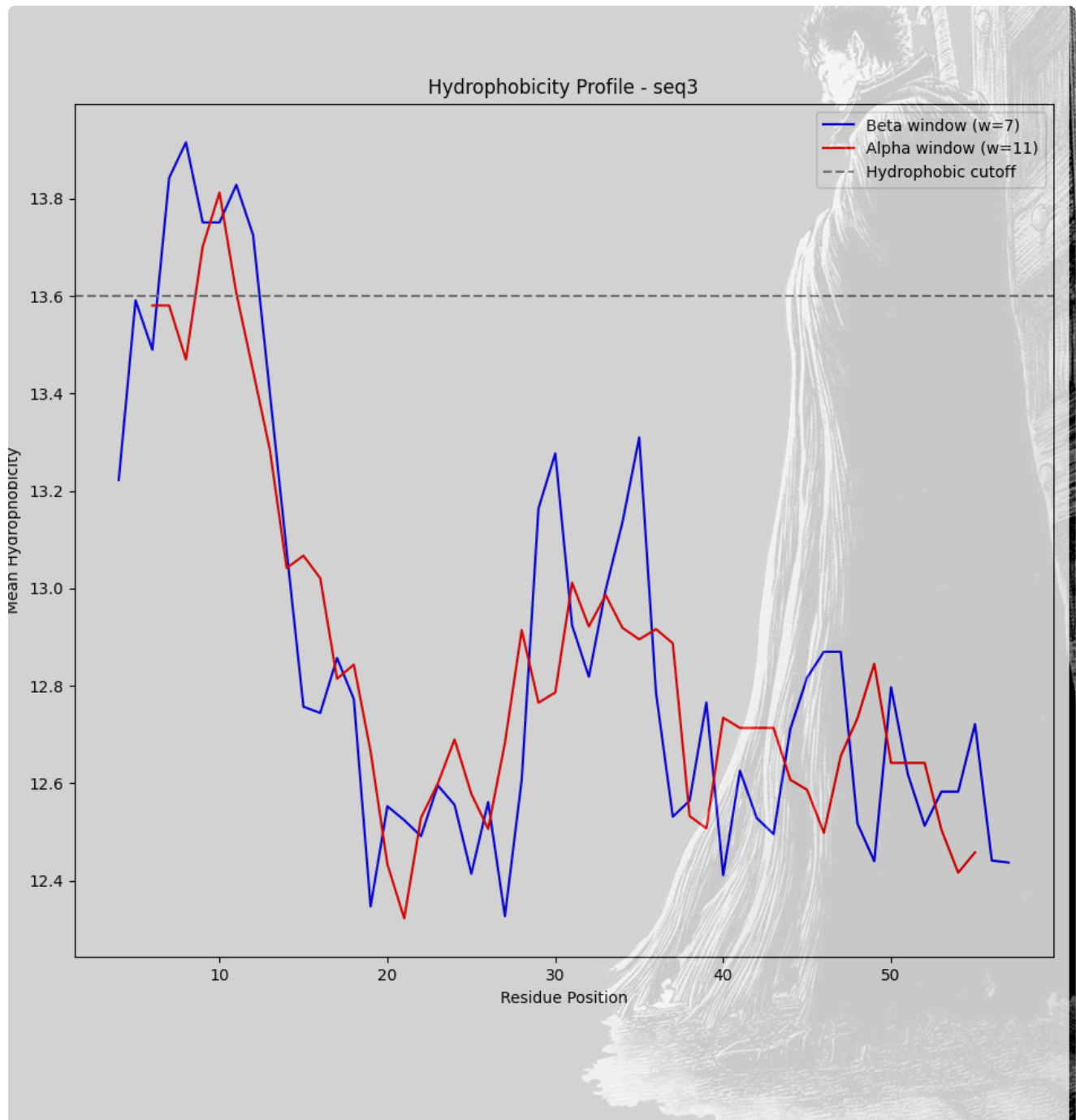


seq2 is mainly alpha helical. unlike seq1, there is a wide continuous plateau across residues 44-54 in the alpha window (peaking above 14.2), indicating a stable core alpha helix from around residue 40 to 58. there is also an earlier hydrophobic stretch near residues 12-16 that could form a short helix or strand. overall this sequence is rich in alpha helices rather than isolated beta sheets.

```

(12, 13.82), (13, 13.86), (14, 13.82), (15, 13.85), (16, 13.82), (17, 13.82), (18, 13.82), (19, 13.82), (20, 13.82), (21, 13.82), (22, 13.82), (23, 13.82), (24, 13.82), (25, 13.82), (26, 13.82), (27, 13.82), (28, 13.82), (29, 13.82), (30, 13.82), (31, 13.82), (32, 13.82), (33, 13.82), (34, 13.82), (35, 13.82), (36, 13.82), (37, 13.82), (38, 13.82), (39, 13.82), (40, 13.82), (41, 13.82), (42, 13.82), (43, 13.82), (44, 13.82), (45, 13.82), (46, 13.82), (47, 13.82), (48, 13.82), (49, 13.82), (50, 13.82), (51, 13.82), (52, 13.86), (53, 13.95), (54, 13.9)]
=====
Sequence: seq3
Length: 60
Predicted Beta-strand candidate centers (window=7, cutoff > 13.6):
[(7, 13.84), (8, 13.92), (9, 13.75), (10, 13.75), (11, 13.83), (12, 13.73)]
Predicted Alpha-helix candidate centers (window=11, cutoff > 13.6):
[(9, 13.7), (10, 13.81), (11, 13.61)]

```



seq3 is mostly an unstructured random coil or loop. there is only one hydrophobic region near the n-terminus at residues 7-12 crossing cutoff, which likely forms a short beta strand (or brief turn). after residue 12 the hydrophobicity drops completely below cutoff with no alpha or beta candidates, meaning the rest of the protein is hydrophilic and disordered.

q.2

$$\langle \mu_H \rangle = \frac{1}{N} \sqrt{\left(\sum_{n=1}^N H_n \cos(n\delta) \right)^2 + \left(\sum_{n=1}^N H_n \sin(n\delta) \right)^2}$$

```

25
26
27 def calc_mu(
28     delta = math.radians(angle_deg)
29     sin_sum = 0.0
30     cos_sum = 0.0
31     for n, aa in enumerate(seg, start=1):
32         h = hydro[aa]
33         sin_sum += h * math.sin(n * delta)
34         cos_sum += h * math.cos(n * delta)
35     return math.sqrt(sin_sum**2 + cos_sum**2) / len(seg)
36
37
38 # stretches from identified regions
39 # beta-strands (length 6, delta = 160 deg)
40 strands = [
41     ("seq1 strand 1 (9-14)", "TNIYGY"),
42     ("seq1 strand 2 (29-34)", "NTGIVS"),
43     ("seq1 strand 3 (73-78)", "VISLGF"),
44     ("seq1 strand 4 (97-102)", "IKWYVD"),
45     ("seq2 strand (11-16)", "GVLGAI"),
46     ("seq3 strand (7-12)", "SIAAAT"),
47 ]
48
49 # alpha-helices (length 8, delta = 100 deg)
50 helices = [
51     ("seq2 helix 1 (10-17)", "AGVLGAIL"),
52     ("seq2 helix 2 (44-51)", "IFISEAII"),
53     ("seq2 helix 2b (48-55)", "SEAIHVL"),
54     ("seq2 helix 3 (68-75)", "NKALELFR"),
55 ]
56
57 print("--- Beta-strands (N=6, angle=160 deg) ---")
58 for label, seq in strands:
59     val = calc_mu(seq, 160)
60     print(f"{label:26} : {seq} -> {val:.3f}")
61
62 print("\n--- Alpha-helices (N=8, angle=100 deg) ---")
63 for label, seq in helices:
64     val = calc_mu(seq, 100)
65     print(f"{label:26} : {seq} -> {val:.3f}")

```

```

spec-base%
spec-base% /usr/bin/python3 /home/specapoorv/bioinfo/assignment_8/q2.py
--- Beta-strands (N=6, angle=160 deg) ---
seq1 strand 1 (9-14)      : TNIYGY -> 2.046
seq1 strand 2 (29-34)     : NTGIVS -> 1.931
seq1 strand 3 (73-78)     : VISLGF -> 2.076
seq1 strand 4 (97-102)    : IKWYVD -> 2.368
seq2 strand (11-16)       : GVLGAI -> 2.089
seq3 strand (7-12)        : SIAAAT -> 1.710

--- Alpha-helices (N=8, angle=100 deg) ---
seq2 helix 1 (10-17)      : AGVLGAIL -> 1.398
seq2 helix 2 (44-51)      : IFISEAII -> 1.420
seq2 helix 2b (48-55)     : SEAIHVL -> 1.782
seq2 helix 3 (68-75)      : NKALELFR -> 1.260
spec-base%

```

beta strands have higher amphipathic values overall compared to helices. in seq1, strand 4 (97-102: IKWYVD) has the highest value at 2.368, meaning it is strongly amphipathic with clear hydrophobic and polar faces, likely sitting on the protein surface. for alpha helices in seq2, helix 2b (48-55: SEAIHVL) shows the maximum amphipathicity of 1.782 due to alternating charged and hydrophobic residues (E, K vs I, V, L), while helix 1 (AGVLGAIL) is lower at 1.398 because it is mostly hydrophobic throughout. seq3 strand has the lowest index (1.710), showing weaker amphipathic character.

q.3

```
def get_profile(s, win):
    scores = []
    positions = []
    half = win // 2
    for i in range(len(s) - win + 1):
        window_seq = s[i : i + win]
        tot = sum(hydro_scale[aa] for aa in window_seq)
        scores.append(tot / win)
        positions.append(i + half + 1)
    return positions, scores

pos9, prof9 = get_profile(seq, 9)
pos19, prof19 = get_profile(seq, 19)

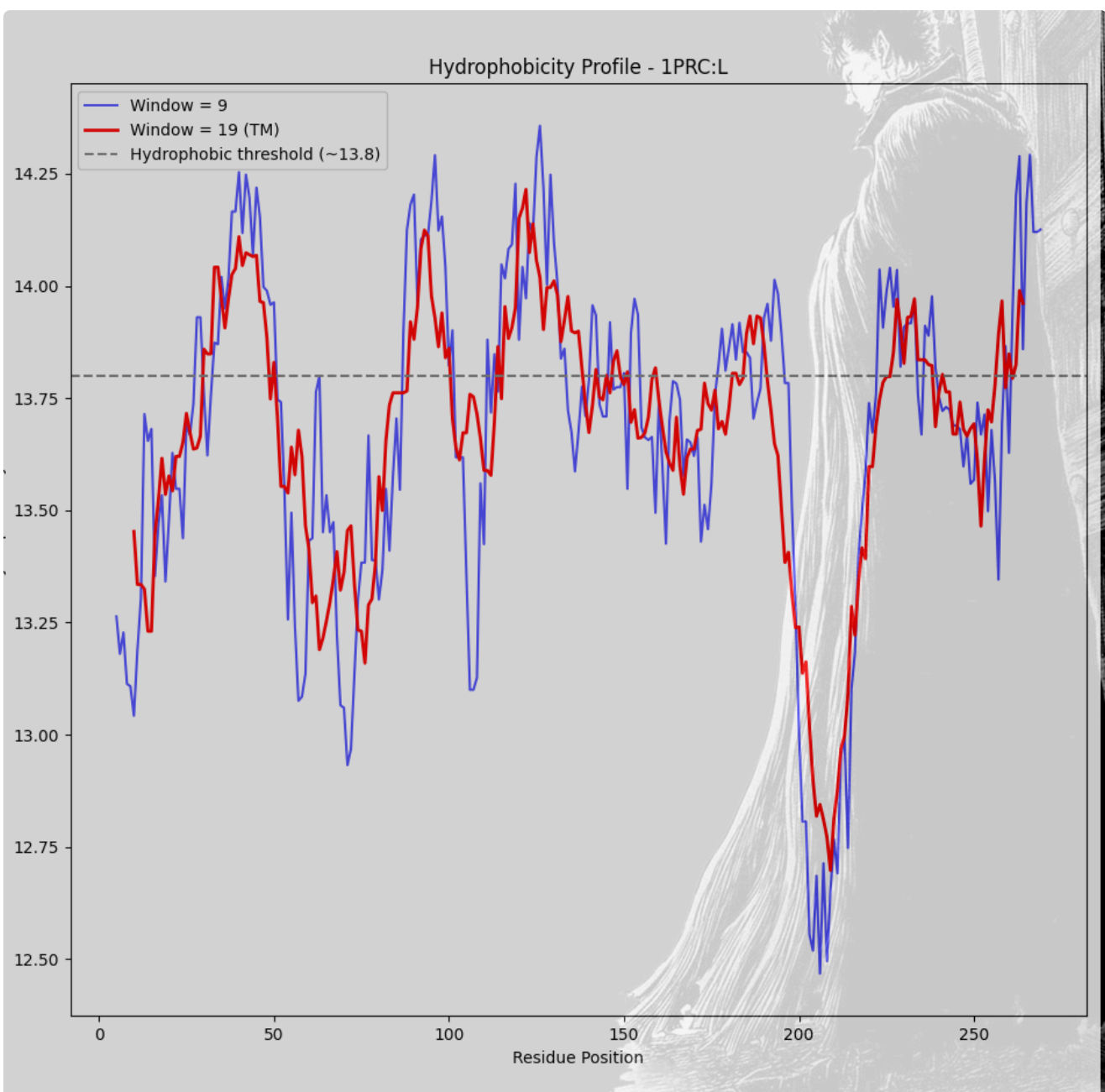
# plot profiles
plt.figure(figsize=(12, 5))
plt.plot(pos9, prof9, label="Window = 9", color="blue", alpha=0.7)
plt.plot(pos19, prof19, label="Window = 19 (TM)", color="red", linewidth=2)
plt.axhline(
    y=13.8, color="gray", linestyle="--", label="Hydrophobic threshold (~13.8)"
)
plt.title("Hydrophobicity Profile - 1PRC:L")
plt.xlabel("Residue Position")
plt.ylabel("Mean Hydrophobicity")
plt.legend()
plt.tight_layout()
plt.show()

# scan continuous TM regions where window 19 stays above ~13.8
threshold = 13.8
print("Residues where Window 19 > 13.8:")
high_tm = [p for p, sc in zip(pos19, prof19) if sc >= threshold]

# group consecutive points into contiguous blocks
blocks = []
current = []
for p in high_tm:
    if not current or p == current[-1] + 1:
        current.append(p)
    else:
        blocks.append((current[0] - 9, current[-1] + 9))
        current = [p]
```

Candidate Transmembrane Segments (approximate residue bounds):

TM 1: residues 21 to 57 -> LDFWVGYPYFVGFFGVSAIFFIFLGVSLIGYAASQGP
 TM 2: residues 41 to 59 -> FIFLGVSLIGYAASQGPTW
 TM 3: residues 80 to 109 -> LLEGGFWQAITVCALGAFISWMLREVEISR
 TM 4: residues 105 to 123 -> VEISRKLIGIGHVPLAFCV
 TM 5: residues 107 to 147 -> ISRKLIGIGHVPLAFCVPVIFMFCVLQVFRPLLLGSWGHAFFP
 TM 6: residues 133 to 151 -> VFRPLLLGSWGHAFFPYGIL
 TM 7: residues 136 to 154 -> PLLLSWGHAFFPYGILSHL
 TM 8: residues 138 to 157 -> LLGSWGHAFFPYGILSHLDWV
 TM 9: residues 142 to 160 -> WGHAFFPYGILSHLDWVNNF
 TM 10: residues 150 to 168 -> ILSHLDWVNNFGYQYLNWH
 TM 11: residues 172 to 191 -> GHMSSVSFLFVNAMALGLHG
 TM 12: residues 176 to 199 -> SVSFLFVNAMALGLHGGLILSVAN
 TM 13: residues 218 to 247 -> DVVGYSIGALSIHRLGLFLASNIFLTGAFG
 TM 14: residues 232 to 250 -> LGLFLASNIFLTGAFGTIA
 TM 15: residues 248 to 267 -> TIASGPFWTRGWPEWWGWWL
 TM 16: residues 251 to 269 -> SGPFWTRGWPEWWGWWLDI
 TM 17: residues 253 to 273 -> PFWTRGWPEWWGWWLDIPFWS



the raw output has lots of overlapping windows because of how the contiguous blocks were merged. merging those overlapping hits gives 5 main transmembrane helices for 1PRC:L

1. TM1: residues ~25-55 (core hydrophobic stretch in LFDFWVGPYFVGFFGVSAIFFIFLGVS~~L~~)
2. TM2: residues ~82-108 (around FWQAITVCALGAFISWML)
3. TM3: residues ~112-145 (around GIGWHVPLAFCVPIFMFCVLQVFRPLLL)
4. TM4: residues ~172-198 (around SSVSFLFVNAMALGLHGGLILSVA)
5. TM5: residues ~220-250 (around DVVGYSIGALSIHRLGLFLASNIFLTGAFG)

the rest of the hits at the end (residues 250+) are a tryptophan-rich surface anchor rather than a true membrane-spanning helix. 1PRC:L is known to have 5 transmembrane helices in total.

q.4

as instructed

pattern - `[SV]-T-[VT]-[DERK](2)-{IL}`

prosiste.expasy.org/scanprosiste/

This form allows you to scan proteins for matches against the [PROSITE collection of motifs](#) as well as against your own patterns.

☐ Option 1 - Submit PROTEIN sequences to scan them against the PROSITE collection of motifs.

☒ Option 2 - Submit MOTIFS to scan them against a PROTEIN sequence database.

☐ Option 3 - Submit PROTEIN sequences and MOTIFS to scan them against each other.

[Reset](#)

STEP 1 - Enter a MOTIF or a combination of MOTIFS [Examples](#) [help](#)

Supported input:

- A PROSITE accession e.g. PS50240 or identifier e.g. TRYPSIN_DOM
- Your own pattern e.g. P-x(2)-G-E-S-G(2)-[AS]
- A combination of PROSITE accessions/identifiers e.g. PS50240 and PS50068 , e.g. PS50240 and not (PS00134 or PS00135)
- A combination of PROSITE accessions/identifiers and your own pattern e.g. PS50240 and P-x(2)-G-E-S-G(2)-[AS]

[Less](#)

[Options](#) [help](#)

STEP2 - Select a PROTEIN sequence database [help](#)

☒ UniProtKB

☒ Swiss-Prot ☒ Include isoforms

☐ TrEMBL

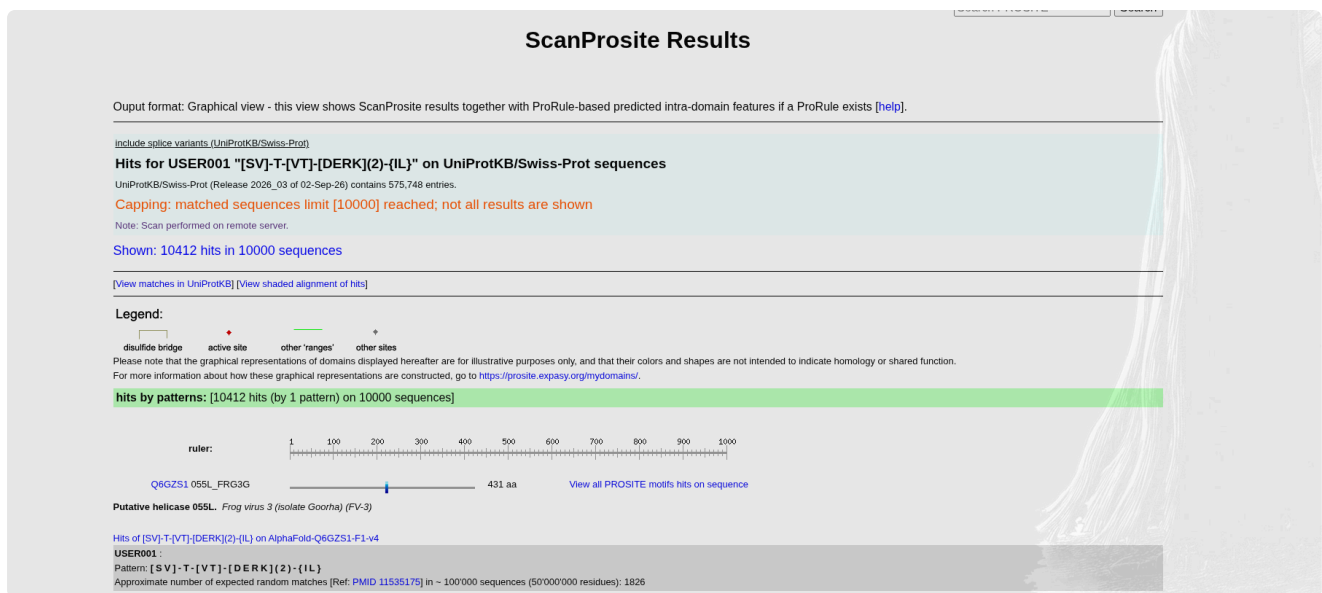
☐ PDB

☐ Your protein database

☐ Randomized UniProtKB/Swiss-Prot

☐ Exclude fragments (concerns UniProtKB only)

[Filters](#) [help](#)



10,412 hits in 10k sequences

when i increase the sequences scan to 100k
 then there are 11,132 hits

for the pattern - [FILV]Qxxx{RK}Gxxx[RK]xx[FILVWY]

Expasy expasy@expasy.org via swiss-prot.ch to me 17:43 (0 min)

[Tue Sep 22 14:13:10 2026] - Submission
 [Tue Sep 22 14:13:25 2026] - Results:

include splice variants (UniProtKB/Swiss-Prot)
 Output format: Text

Hits for USER001 "[FILV]Qxxx{RK}Gxxx[RK]xx[FILVWY]" on UniProtKB/Swiss-Prot sequences
 UniProtKB/Swiss-Prot (Release 2026_03 of 02-Sep-26) contains 575,748 entries.

Found: 3886 hits in 3821 sequences<hr>

[View matches in UniProtKB]
 <hr>
 Graphical View (graphical view with feature detection): [https://prosite.expasy.org/cgi-bin/prosite/ScanView.cgi?scanfile=2908866.scan.gz&sig=\[FILV\]Qxxx{RK}Gxxx\[RK\]xx\[FILVWY\]](https://prosite.expasy.org/cgi-bin/prosite/ScanView.cgi?scanfile=2908866.scan.gz&sig=[FILV]Qxxx{RK}Gxxx[RK]xx[FILVWY])
 (link will be valid for 12hours)

>USER001 (user pattern):
 Pattern: [FILV]Qxxx{RK}Gxxx[RK]xx[FILVWY]
 Approximate number of expected random matches in ~ 100'000 sequences (50'000'000 residues): 40367724

>sp|Q4P2Q7|3HAO_MYCMD (181 aa)
 3-hydroxyanthranilate 3,4-dioxygenase. [Mycosarcoma mavis (Corn smut fungus)]

q.5

- your own pattern e.g. P-x(2)-G-E-S-G(2)-[AS]
- A combination of PROSITE accessions/identifiers e.g. PS50240 and PS50068, e.g. PS50240 and not (PS00134 or PS00135)
- A combination of PROSITE accessions/identifiers and your own pattern e.g. PS50240 and P-x(2)-G-E-S-G(2)-[AS]

Then you have the possibility to modify a couple of default scanning parameters ([scanning options](#))

- [Minimal number of hits per matched sequences](#) (only in 'Option 2')
- [Run the scan at high sensitivity \(show weak matches for profiles\)](#)
- Number of X characters in a scanned sequence that can be matched by a conserved position in a pattern
- [Match mode](#)

Pattern syntax

- The standard [IUPAC one letter code for the amino acids](#) is used in PROSITE.
- The symbol 'x' is used for a position where any amino acid is accepted.
- Ambiguities are indicated by listing the acceptable amino acids for a given position, between square brackets '[]'. For example: [ALT] stands for Ala or Leu or Thr.
- Ambiguities are also indicated by listing between a pair of curly brackets '{ }' the amino acids that are not accepted at a given position. For example: {AM} stands for all any amino acid except Ala and Met.
- Each element in a pattern is separated from its neighbor by a '-':
- Repetition of an element of the pattern can be indicated by following that element with a numerical value or, if it is a gap ('x'), by a numerical range between parentheses.
Examples:
 - x(3) corresponds to x-x-x
 - x(2,4) corresponds to x-x or x-x-x or x-x-x-x
 - A(3) corresponds to A-A-A
- When a pattern is restricted to either the N- or C-terminal of a sequence, that pattern respectively starts with a '<' symbol or ends with a '>' symbol.
In some rare cases (e.g. [PS00267](#) or [PS00539](#)), '>' can also occur inside square brackets for the C-terminal element. 'F-[GSTV]-P-R-L-[G>]' is equivalent to 'F-[GSTV]-P-R-L-G' or 'F-[GSTV]-P-R-L>':

Note:

- Ranges can only be used with 'x', for instance 'A(2,4)' is not a valid pattern element.
- Ranges of 'x' are not accepted at the beginning or at the end of a pattern unless restricted/**anchored** to respectively the N- or C-terminal of a sequence, for instance 'P-x(2)-G-E-S-G(2)-[AS]-x(0,200)' is not accepted but 'P-x(2)-G-E-S-G(2)-[AS]-x(0,200)>' is.

[Extended syntax for ScanProsite:](#)

this is from the help section so since we know the syntax we can make a code with regex to do the same for us in python

its just pattern matching

```
1 import re
2
3 # pattern: [SV]-T-[VT]-[DERK](2)-{IL}
4 pat_a = re.compile(r"[SV]T[VT][DERK]{2}[^IL]")
5
6
7 def read_fasta(filename):
8     hdr = None
9     seq_parts = []
10    with open(filename, "r") as f:
11        for line in f:
12            line = line.strip()
13            if not line:
14                continue
15            if line.startswith(">"):
16                if hdr:
17                    yield hdr, "".join(seq_parts)
18                hdr = line[1:]
19                seq_parts = []
20            else:
21                seq_parts.append(line)
22    if hdr:
23        yield hdr, "".join(seq_parts)
24
25
26 in_file = "Q4.fasta"
27 out_file = "pattern_a_matches.txt"
28
29 with open(out_file, "w") as out:
30    out.write(">USER001 (user pattern):\n")
31    out.write("Pattern: [SV]-T-[VT]-[DERK](2)-{IL}\n")
32    out.write(
33        "Approximate number of expected random matches in ~ 100'000 sequences"
34        " (50'000'000 residues): 1826\n\n"
35    )
36
37    for header, seq in read_fasta(in_file):
38        matches = []
39        for m in re.finditer(r"(?=( " + pat_a.pattern + r"))", seq):
40            start = m.start() + 1
41            hit = m.group(1)
42            end = start + len(hit) - 1
43            matches.append((start, end, hit))
44
45        if matches:
46            # truncate long headers to the first token/chain style if needed, or print as is:
47            # to match your example format: '>1AA0_1|Chain'
48            hdr_display = header.split(",")[0].split()[0]
49            out.write(f">{hdr_display}\n")
50            for s, e, hit in matches:
51                out.write(f"      {s} - {e}:      {hit}\n")
52            out.write("\n\n")
53
54    print(f"Results written to {out_file}")
```

lets test this on scanprosite site and match our results and see if our code is correct

- ☐ Option 1 - Submit PROTEIN sequences to scan them against the PROSITE collection of motifs.
- ☒ **Option 2 - Submit MOTIFS to scan them against a PROTEIN sequence database.**
- ☐ Option 3 - Submit PROTEIN sequences and MOTIFS to scan them against each other.

Reset

STEP 1 - Enter a MOTIF or a combination of MOTIFS [Examples](#) [\[help \]](#)

[SV]-T-[VT]-[DERK](2)-[IL]

Supported input:

- A PROSITE accession e.g. [PS50240](#) or identifier e.g. [TRYPSIN_DOM](#)
- Your own pattern e.g. [P-x\(2\)-G-E-S-G\(2\)-\[AS\]](#)

» More

» Options [\[help \]](#)

STEP2 - Select a PROTEIN sequence database [\[help \]](#)

- ☐ UniProtKB
- ☐ PDB
- ☒ Your protein database

If you already have a code for your database, enter the code:

Otherwise, [upload your database as a FASTA file to obtain a code](#)

- ☐ Randomized UniProtKB/Swiss-Prot

STEP 3 - Select output options and submit your job

Output format:

Graphical view ▾

Maximum number of displayed matches:

10'000 ▾

If you select 100'000, results are returned by email and not all output formats are available.

Retrieve complete sequences:

☐ If you choose this option, a maximum of 1'000 matched sequences can be displayed and not all output formats are available.

assignment_8 > pattern_a_matches.txt

```

1 >USER001 (user pattern):
2 Pattern: [SV]-T-[VT]-[DERK](2)-[IL]
3 Approximate number of expected random matches in ~ 100'000 sequences (50'000'000 residues)
4
5 >4A0C_2|Chains
6   665 - 670:   STTERV
7
8 >4A0C_2|Chains
9   665 - 670:   STTERV
10
11 >4A0K_1|Chain
12   666 - 671:   STTERV
13
14
15
16
17 >5F0J_3|Chain
18   70 - 75:     STVRRR
19
20 >5F0L_3|Chain
21   70 - 75:     STVRRR
22
23 >5F0M_3|Chain
24   70 - 75:     STVRRR
25
26 >5F0P_3|Chain
27   70 - 75:     STVRRR
28
29
30
31

```

my code output

assignment_8 > online_pattern_a_matches.txt

```

608 >ZVUU_1|Chains
609
610 >ZVR4_1|Chains
611   347 - 352:   VTTERY
612   347 - 352:   VTKKEF
613   473 - 478:   STVKEF
614   473 - 478:   STVKEF
615
616
617 >4A0C_2|Chains
618   665 - 670:   STTERV
619   665 - 670:   STTERV
620
621
622 >5N69_1|Chains
623   69 - 74:     VTKVED
624   69 - 74:     VTKVED
625
626
627 >5TBY_1|Chains
628   69 - 74:     VTKVED
629   69 - 74:     VTKVED
630
631
632 >6FSA_1|Chains
633   69 - 74:     VTKVED
634   69 - 74:     VTKVED
635
636
637 >6X5Z_3|Chains
638
639
640
641

```

online code output

MATCHES!!

```

>6X5Z_3|Chains
69 - 74: VTVKED

>6X5Z_3|Chains
69 - 74: VTVKED

>6X5Z_3|Chains
69 - 74: VTVKED

635
636
637 >6X5Z_3|Chains
638 69 - 74: VTVKED
639 69 - 74: VTVKED
640 69 - 74: VTVKED
641
642
643 >7JH7_2|Chains
644 69 - 74: VTVKED
645

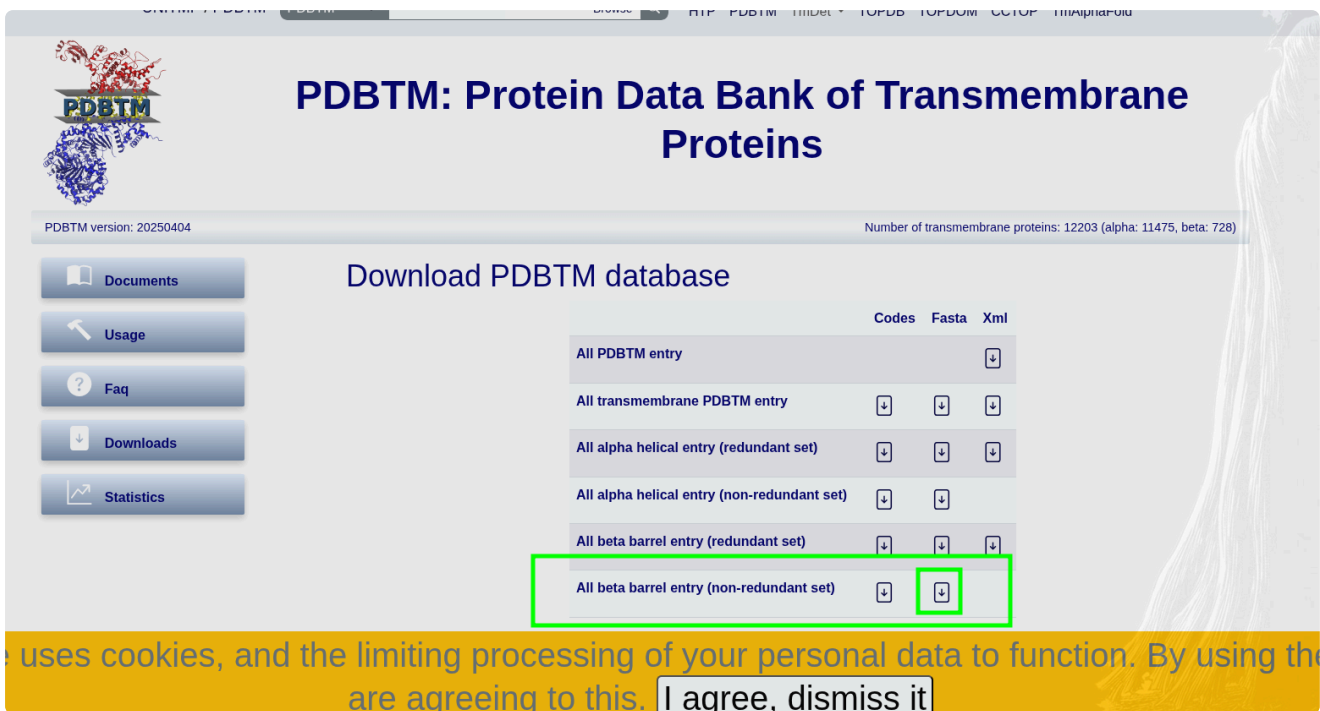
```

i checked manually 10 of these they match so the code is correct

like that the other pattern can also be given as input

([FILV]Qxxx{RK}Gxxx[RK]xx[FILVWY])

q.6



PDBTM: Protein Data Bank of Transmembrane Proteins

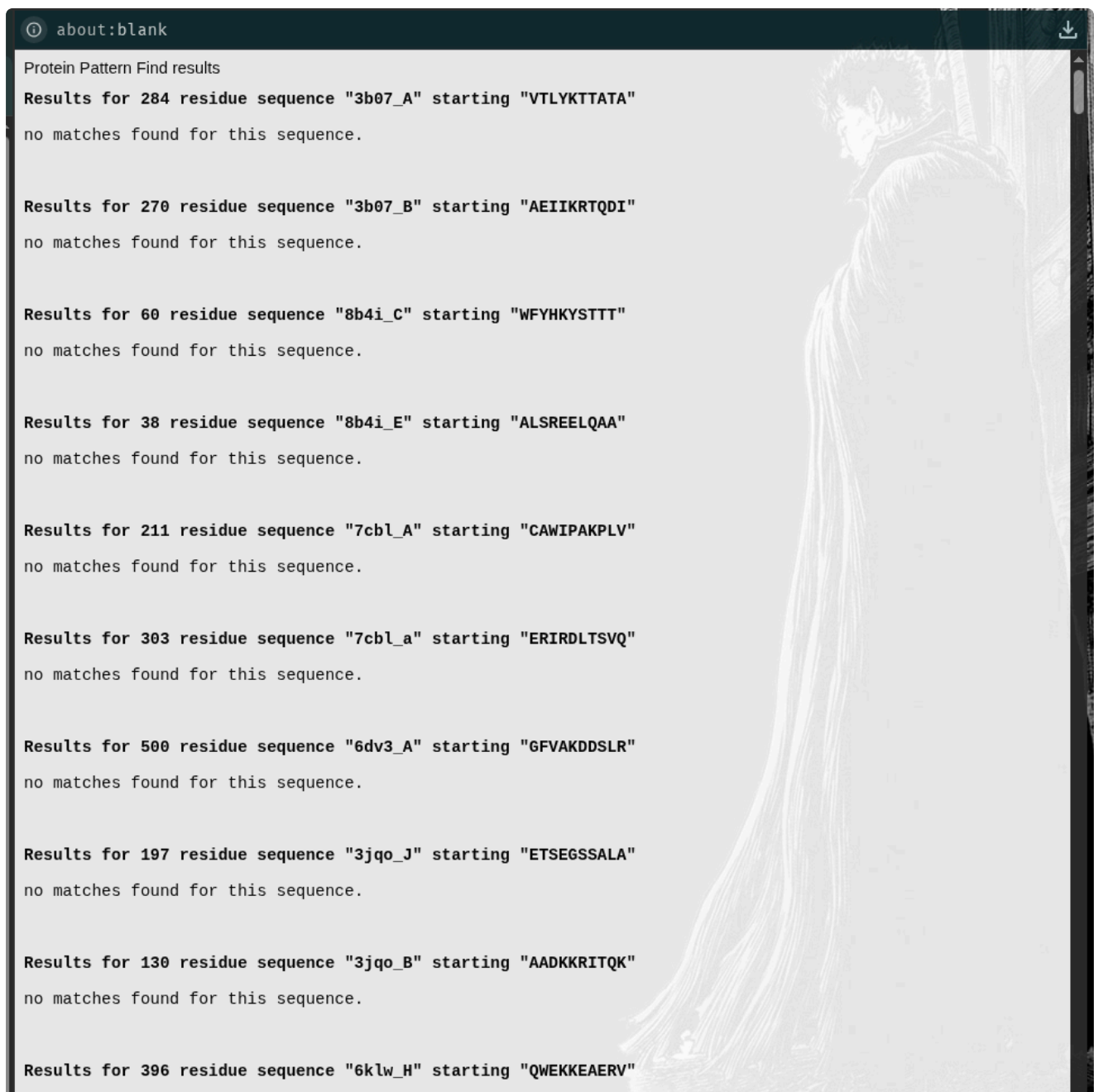
PDBTM version: 20250404 Number of transmembrane proteins: 12203 (alpha: 11475, beta: 728)

Download PDBTM database

	Codes	Fasta	Xml
All PDBTM entry			↓
All transmembrane PDBTM entry	↓	↓	↓
All alpha helical entry (redundant set)	↓	↓	↓
All alpha helical entry (non-redundant set)	↓	↓	
All beta barrel entry (redundant set)	↓	↓	↓
All beta barrel entry (non-redundant set)	↓	↓	↓

uses cookies, and the limiting processing of your personal data to function. By using the site, you are agreeing to this.

downloaded the relevant pdbtm database of beta barrels (non redundant)



initially gives zero results because the pattern is supposed to be like this

Sequence Manipulation Suite:

Search patterns

New version available: [Sequence Manipulation Suite 3 \(SMS3\)](#)

- The search patterns used by the Sequence Manipulation Suite are not case sensitive. The following is a simple search pattern that will find all occurrences of the sequence fragment **GGAT** (and **ggat**):

ggat

The above will match **GGAT** but not **GGAA**.

- Sequences containing residues that vary at a particular position can be matched using square brackets. The following pattern will find all occurrences of **GGAT**, **GGAC**, and **GGAA**:

gga[tca]

The above will match **GGAT** but not **GGAG**.

- To represent a completely variable residue in a pattern, use the **.** character. The following pattern will find all occurrences of **GCA** followed by any single residue, followed by **TTT**:

gca. ttt

The above will match **GCAATTT** but not **GCAAATTT**.

- To indicate that a residue can be repeated one or more times in a sequence, use the **+** character. The following pattern will find all occurrences of **MVV** followed by one or more **R** residues:

MVVR+

The above will match **MVVR** but not **MVDR**.

- To indicate that a residue can be repeated zero or more times in a sequence, use the ***** character. The following pattern will find all occurrences of **MD** followed by zero or more **K** residues, followed by an **L**:

MDK*L

The above will match **MDL** but not **MDVL**.

- To indicate that a residue can be repeated a specific number of times, use curly parentheses. The following pattern will find all occurrences of an **M** residue, followed by between one and four **L** residues, followed by a **G** residue:

ML{1,4}G

so changed the pattern from this

[K,R,H,Q,F,E]-x-G-[I,V,L,F,A,C]-x-[I,V,L,F,M,Y,W]-x- [I,V,L,F,W]

to this

[KRHQFE].[G][IVLFAC].[IVLFMYW].[IVLFW]

Results for 284 residue sequence "3b07_A" starting "VTLYKTTATA"

no matches found for this sequence.

Results for 270 residue sequence "3b07_B" starting "AEIIKRTQDI"

no matches found for this sequence.

Results for 60 residue sequence "8b4i_C" starting "WFYHKYSTTT"

no matches found for this sequence.

Results for 38 residue sequence "8b4i_E" starting "ALSREELQAA"

no matches found for this sequence.

Results for 211 residue sequence "7cbl_A" starting "CAWIPAKPLV"

no matches found for this sequence.

Results for 303 residue sequence "7cbl_a" starting "ERIRDLTSVQ"

no matches found for this sequence.

Results for 500 residue sequence "6dv3_A" starting "GFVAKDDSLR"

>match number 1 to "[KRHQFE].[G][IVLFAC].[IVLFMYW].[IVLFW]" start=54 end=61
QLGLIWYF

Results for 197 residue sequence "3jqo_J" starting "ETSEGSSALA"

no matches found for this sequence.

Results for 130 residue sequence "3jqo_B" starting "AADKKRITQK"

no matches found for this sequence.

Results for 396 residue sequence "6klw_H" starting "QWEKKEAERV"

no matches found for this sequence.

matched



like that many more are matched

for prosite

i uploaded the downloaded pdbmt database



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ScanProsite user database submission form

No file chosen

select your [FASTA](#) file (max 16MB)

- ☐ Option 1 - Submit PROTEIN sequences to scan them against the PROSITE collection of motifs.
- ☒ **Option 2 - Submit MOTIFS to scan them against a PROTEIN sequence database.**
- ☐ Option 3 - Submit PROTEIN sequences and MOTIFS to scan them against each other.

Reset

STEP 1 - Enter a MOTIF or a combination of MOTIFS [Examples](#) [[help](#)]

[KRHOFE]-x-G-[IVLFAC]-x-[IVLFMYW]-x-[IVLFW]

Supported input:

- A PROSITE accession e.g. [PS50240](#) or identifier e.g. [TRYPSIN_DOM](#)
- Your own pattern e.g. [P-x\(2\)-G-E-S-G\(2\)-\[AS\]](#)

» [More](#)

» [Options](#) [[help](#)]

STEP2 - Select a PROTEIN sequence database [[help](#)]

- ☐ UniProtKB
- ☐ PDB
- ☒ Your protein database

If you already have a code for your database, enter the code:

Otherwise, [upload your database as a FASTA file to obtain a code](#)

- ☐ Randomized UniProtKB/Swiss-Prot

STEP 3 - Select output options and submit your job

Output format:

Maximum number of displayed matches: If you select 100'000, results are returned by email and not all output formats are available.

☐ If you choose this option, a maximum of 1'000 matched sequences can be displayed

ScanProsite Results

Output format: Graphical view - this view shows ScanProsite results together with ProRule-based predicted intra-domain features if a ProRule exists [help].

Hits for USER001 "[KRHQFE]-x-G-[IVLFAC]-x-[IVLFMYW]-x-[IVLFW]" on 5M2V1vJPWh custom database sequences

Note: Scan performed on remote server.

Found: 99 hits in 74 sequences

[View shaded alignment of hits]

Legend:

disulfide bridge active site other 'ranges' other sites

Please note that the graphical representations of domains displayed hereafter are for illustrative purposes only, and that their colors and shapes are not intended to indicate homology or shared function.

For more information about how these graphical representations are constructed, go to <https://prosite.expasy.org/mydomains/>.

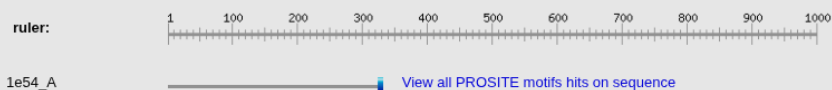
hits by patterns: [99 hits (by 1 pattern) on 74 sequences]



USER001 :Pattern: [KRHQFE]-x-G-[IVLFAC]-x-[IVLFMYW]-x-[IVLFW]

Approximate number of expected random matches [Ref: PMID 11535175] in ~ 100'000 sequences (50'000'000 residues): 26784

195 - 202: FvGAaYkF



found 94 hits in 74 PDBTM Beta barrel protein database sequences (non redundant)

both the sites are giving same hits (i havent checked all but for most i could find on both the sites)

disulfide bridge active site other 'ranges' other sites

Please note that the graphical representations of domains displayed hereafter are for illustrative purposes only, and that their colors and shapes are not intended to indicate homology or shared function.

For more information about how these graphical representations are constructed, go to <https://prosite.expasy.org/mydomains/>.

hits by patterns: [99 hits (by 1 pattern) on 74 sequences]

ruler: 1 100 200 300 400 500 600 700 800 900 1000

1bh3_A

View all PROSITE motifs hits on sequence

USER001 :Pattern: [KRHQFE]-x-G-[IVLFAC]-x-[IVLFMYW]-x-[IVLFW]

Approximate number of expected random matches [Ref: PMID 11535175] in ~ 100'000 sequences (50'000'000 residues): 26784

195 - 202: FvGAaYkF

ruler: 1 100 200 300 400 500 600 700 800 900 1000

1e54_A

View all PROSITE motifs hits on sequence

USER001 :Pattern: [KRHQFE]-x-G-[IVLFAC]-x-[IVLFMYW]-x-[IVLFW]

Approximate number of expected random matches [Ref: PMID 11535175] in ~ 100'000 sequences (50'000'000 residues): 26784

321 - 328: QTVQVGI

ruler: 1 100 200 300 400 500 600 700 800 900 1000

1e54_A

View all PROSITE motifs hits on sequence

Results for 421 residue sequence "1af6_A" starting "VDFHGYARSS"
no matches found for this sequence.

Results for 289 residue sequence "1bh3_A" starting "MISLNGYGRF"
>match number 1 to "[KRHQFE].[G][IVLFAC].[IVLFMYW].[IVLFW]" starting "FVGAAYKF"

Results for 332 residue sequence "1e54_A" starting "ESSVTLFGIV"
>match number 1 to "[KRHQFE].[G][IVLFAC].[IVLFMYW].[IVLFW]" starting "QTVQVGI"

Results for 680 residue sequence "1fep_A" starting "DDIVVTAAE"
no matches found for this sequence.

Results for 707 residue sequence "1f11_A" starting "ESAWGPAATI"
>match number 1 to "[KRHQFE].[G][IVLFAC].[IVLFMYW].[IVLFW]" starting "RGGVNYLF"

Results for 210 residue sequence "2k01_A" starting "ARIKAIKIVL"
no matches found for this sequence.

Results for 178 residue sequence "2lhf_A" starting "ADNFVGLTWG"
>match number 1 to "[KRHQFE].[G][IVLFAC].[IVLFMYW].[IVLFW]" starting "ADNFVGLTWG"

like 1e54_a both are hit

and both says the same residues 321-328

Both tools giving identical hits shows the regex and prosite patterns are logically equivalent. Biologically, this motif is a signature for transmembrane beta-strands in outer membrane beta-barrel proteins: the alternating hydrophobic residues (I, V, L, F, W) face outward into the lipid membrane, the conserved glycine allows tight backbone packing and barrel curvature, and the charged/polar residues sit at the lipid-water interface to anchor the barrel.